

Protocols are systems: blueprint

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Chapter 1

Systems and the floor

Definition 1.1 (System). A system is a pair $S = (T, R)$ of a set of things T and a relation $R \subseteq T \times T$.

Remark. Klir, *Facets of Systems Science* (2001), p. 5: “S, T, R denote, respectively, a system, a set of things distinguished within S, and a relation (or, possibly, a set of relations) defined on T. Clearly, the thinghood and systemhood properties of S reside in T and R, respectively.” His examples on the same page admit both an ordering, which runs one way, and a partition, which comes from an equivalence relation and runs both ways.

Definition 1.2 (Dependency). X depends on Y when what X is cannot be stated without Y . For the parts of Definition 1.1 this is a matter of type: R is a subset of $T \times T$, so the kind of object R is depends on T . That R depends on T , and T not on R , holds by construction.

Definition 1.3 (Shape of a definition). The shape $\mathcal{J}_{\text{Klir}}$ has two objects, the relation and the things, and one generating arrow

$$R \longrightarrow T,$$

read “ R is defined over T ”. The arrow joins the parts of the definition, not two things.

Remark (Data and dependency). Klir’s data form a graph. Each element of R is a pair with a first and a second thing, so R maps to T twice, $R \rightrightarrows T$, and read as a schema for data, that two-arrow shape is the shape of graphs. The one-arrow shape is not a schema for data. It records dependency: collapsing both arrows to one forgets *how* R refers to T and keeps only *that* it does.

$$\underbrace{R \rightrightarrows T}_{\text{data}} \rightsquigarrow \underbrace{R \longrightarrow T}_{\text{dependency}}$$

The comparison across definitions takes place at the dependency level. The two-arrow shape is not shared: the Myers shape has at most one arrow between any two positions, so two parallel arrows have nowhere to go. The collapse is an explanation and is not formalized.

Definition 1.4 (Kernel). A kernel is a system (T, R) together with the constraint that every pair in R lies in T . The constraint is the arrow of Definition 1.3 at the level of data.

Theorem 1.5 (The arrow is in every definition). *For each of the definitions of Bunge, Mobus, Myers, Wymore, Mesarović, Joslyn, and Spivak, there is a functor from $\mathcal{J}_{\text{Klir}}$ to that definition’s shape that is injective on objects. With Klir’s own, the arrow sits inside all eight.*

Proof. Each functor is given by a choice of positions in the target shape. Faithfulness is automatic because $\mathcal{J}_{\text{Klr}}$ is thin, so the content is injectivity on objects. $\square$

Remark (Position, not meaning). The functors place the arrow by position. They do not show that it means the same thing in each definition. Into Myers it lands on **expose**, by which state determines output, a functional dependence. Into Joslyn it lands on **efferent**, the effector acting on the controlled variables, a causal one. The shape files also declare different arrow conventions. Bunge's draws from dependent to depended-on. Joslyn's, Spivak's, Wymore's, Mesarović's, and Willems's draw in the functional direction. Mobus's declares none, and Myers's is inconsistent. See Problem 2.4.

Lemma 1.6 (No two arrows in a row). *No quiver that embeds in the Willems shape contains a path of two arrows.*

Proof. The Willems shape has no composable pair of arrows. $\square$

Theorem 1.7 (Only one shared arrow). *Let V be a connected quiver that embeds in both the Joslyn and the Willems shapes, and let $e : x \rightarrow y$ be an edge of V . Then $x \neq y$, $V = \{x, y\}$, and every edge of V runs from x to y .*

Proof. Joslyn's shape forbids two arrows leaving one object, and Willems's forbids two arrows entering one object and any two arrows in a row (Lemma 1.6). A connected quiver satisfying all three has one edge. $\square$

Remark. The result is relative to the encoded presentations. Willems serves as a witness and is not one of the eight. Among things, a relation may run both ways. Among the parts of a definition, the shared floor is one arrow.

Theorem 1.8 (Nothing larger is shared, refuted). *Compared as free categories, the three-object fork embeds injectively on objects and faithfully into all eight shapes. So the claim that nothing larger than one arrow is shared is false at that level.*

Proof. In a free category every path is a morphism, and the fork enters through composite paths that no definition asserts. $\square$

Chapter 2

What the floor can and cannot carry

Theorem 2.1 (Views are generated from the kernel). *A kernel with at least one bonded pair of distinct things generates a Bunge system, and a kernel in which nothing is related to itself generates a Mobus system. Each generated system projects back to the same kernel.*

Proof. The elaboration slots are filled with minimal witnesses: empty environment, no external flows, trivial boundary, history, and timescale. The relation passes through unchanged. $\square$

Theorem 2.2 (The generated energy view is static). *The Spivak system generated from any kernel is static. Motion needs value data from outside the kernel.*

Proof. The generated value type is trivial, so the potential distinguishes no states and the step leaves every state fixed. $\square$

Theorem 2.3 (A one-way relation has no Rosen view). *If some pair in a kernel's relation lacks its converse, no Rosen system (Rosen 1978) projects to that kernel. Rosen's relation among states is always symmetric.*

Proof. A Rosen system's relation is indistinguishability by observables, which is symmetric. $\square$

Problem 2.4 (What the shared arrow means). Put all eight shapes in one declared arrow convention and check whether Theorems 1.5 and 1.7 still hold. Then take dependency in the sense of Definition 1.2 as the weakest notion and ask whether functional dependence (Myers) and causal dependence (Joslyn) refine it. If they do, the shared arrow has one meaning, refined differently by each tradition.

Problem 2.5 (Time and environment). The kernel holds places for time and environment but leaves them empty. Find which time and environment content, if any, can be generated from the kernel, and state exactly what must be supplied from outside.

Problem 2.6 (Emergence). Bunge, *Treatise on Basic Philosophy*, vol. 4 (1979), Definitions 1.13 and 1.14, defines emergent properties set-theoretically. State emergence over the kernel without building the emergent structure into the model.

Chapter 3

Protocols

Definition 3.1 (Protocol). A protocol is a relation R over a set T of roles: what each role does relative to the others. This is a working definition and is not yet formalized.

Definition 3.2 (Protocol system). A protocol system is a protocol together with participants acting in relation to it. Walch, *The Fundamentals of Protocol Systems* (2023), p. 5: “A protocol system is made up of a protocol, at least two people, and those people taking actions in relation to the protocol.” This is a working definition and is not yet formalized.

Definition 3.3 (Layers). Let T be a set of roles and P a set of participants. A protocol system has three layers.

- (L0) Definition: a relation $R \subseteq T \times T$.
- (L1) Occupancy: an assignment $a: P \rightarrow T$ of participants to roles.
- (L2) Enactment: actions by participants, over time, in relation to R .

A protocol is (L0). A protocol system is all three. With arrows from dependent to depended-on,

$$R \xrightarrow{\text{is defined over}} T \xleftarrow{\text{occupies}} P.$$

(L2) has no arrow. It is time, which the kernel holds a place for and cannot fill (Theorem 2.2).

Remark. Three definitions of protocol, read by layer. Walch (2023): “rules created by humans that shape the behavior of groups of people” (p. 4) at (L0), “at least two people” at (L1), and “taking actions in relation to” the protocol at (L2) (p. 5). Rao, Beiko, Ryan, Stark, Van Epps, and Aue, *The Unreasonable Sufficiency of Protocols* (2023), p. 3: a “codified set of behaviors” at (L0) that, “when adopted by a sufficient number of participants” (L1), “reliably leads to good-enough outcomes for all” (L2). Tay, *A Phenomenology of Protocols* (2023), p. 3: “rules or procedures” at (L0), with participants and protocols that “co-constitute reality” (p. 10) at (L2). Every source puts the protocol at (L0). Only Walch names the protocol system as a separate thing.

Problem 3.4 (Unadopted protocols). Does a protocol exist before anyone adopts it? Rao et al. (p. 7): a protocol that falls short “arguably ... wouldn’t exist at all”. Yet they describe IPv6 as ratified in 2017 with adoption still slow (p. 16). Position: yes. R needs roles, not occupants, so what fails to get established is the protocol system, at (L1) and (L2).

Problem 3.5 (Where systemhood lives). Klir places systemhood in R . Walch (p. 3): “Protocol systems are always two things at once—individuals and a group.” Position: both hold, at different layers. Organization lives at (L0). The two-sidedness lives at (L1): the same participants, seen through a , are individuals in roles, and seen through R , are a group.

Problem 3.6 (Do rules make roles?). Some rules create the activity they govern: there is no checkmate before the rules of chess. If a role is made by its rules, then T depends on R and the arrow reverses. Position: a role has a label, which tells the position apart, and a function, what its holder does. The label belongs to T (Walch, p. 6: a role “has a name that distinguishes it from other roles within the system”). The function is R . Rules make labels into roles. A piece carved as a bishop and assigned no moves is $(T, \emptyset)$. “Moves diagonally” cannot be stated without saying what moves.

Problem 3.7 (Formalize protocols). Define a protocol as a relation over roles and a protocol system as an assignment of participants to those roles, and prove that every protocol system is a kernel.

Chapter 4

Three instances

Definition 4.1 (Contacts since the last wash). With $N = 3$, one contact updates the count $c \in \mathbb{N}$ by

$$f(c) = \begin{cases} 0 & \text{if } c + 1 \geq N, \\ c + 1 & \text{otherwise.} \end{cases}$$

Theorem 4.2 (Hand washing keeps contamination bounded). *For all $n, c \in \mathbb{N}$, if $c \leq N$ then $f^n(c) \leq N$.*

Proof. Induction on n , using that one step never exceeds N . □

Remark. The model is coarse: hands and the rule, against surfaces and microbes. It assumes the rule is followed.

Definition 4.3 (Organized). A set S is organized when $\exists a, b \in S$ with $a \neq b$ and $\text{Bonded}(a, b)$. Two systems with disjoint compositions and at least one bond between them compose by union.

Theorem 4.4 (The TCP and HTTP stack is organized). $\text{Org}(C_{\text{HTTP}} \cup C_{\text{TCP}})$, where the bond is the HTTP client handing data to the TCP sender.

Proof. Composition of two systems with a cross bond is organized. □

Remark. Each layer is modeled as closed, with no network environment. This is composition of two systems, not composition of arrows.

Definition 4.5 (Difficulty retarget as a homeostat). The state is the inter-block time $s \in \mathbb{Z}$ in minutes, with set point 10. The feedback law is

$$\varphi(s) = s - (s - 10).$$

Theorem 4.6 (The ten-minute target is a fixed point). $\varphi(10) = 10$.

Proof. At the set point the error is zero and the correction leaves the state unchanged. □

Remark. This is an idealized one-step model of the 2016-block retarget. As written, φ returns 10 from any state, so the result concerns the homeostat's structure, not Bitcoin's dynamics. The real rule adjusts difficulty multiplicatively over a 2016-block window.

Chapter 5

Open problems

Problem 5.1 (Measuring hardness). Hardness can be located in a protocol by drawing its dependencies. Find a measure of hardness that compares protocols of different kinds.

Problem 5.2 (A fifth formalism). State machines, process calculi, temporal logics, and Petri nets can be drawn over a common skeleton. Encode a fifth formalism and test whether the skeleton survives.

Problem 5.3 (Designed hardness). Blygger replaces deletion with withdrawal and makes pins irrevocable. Model it as an instance and locate its hardness.